CCCA Cell-State Analysis — Wang et al. 2019, Glioblastoma

Brain · Glioblastoma · 10x · 10 patients / 16,317 cells (master) · 9,634 malignant cells analysed

GO / NO-GO GATES

Gate 1 — mixing & continuity

GO

7/10 samples (≥237 cells) intermix on one continuous manifold; no per-patient islands (16 detached cells removed); Harmony drops same-sample neighbour fraction 0.99→0.63, iLISI 1.01→2.00.

Gate 2 — module geography

GO (non-branching)

Single developmental axis, not a bifurcation: Gast & pSom co-vary weakly (r=+0.15), Crest is the inverse terminal gradient (vs pSom r=−0.87; Crest-high decile at pseudotime 0.86). No competing arms.

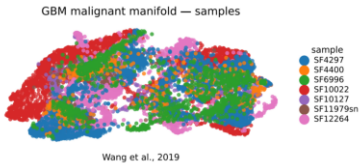
Gate 3 — hub cross-repression

NA

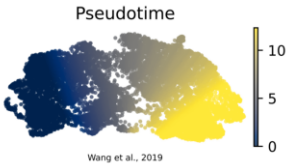
Branching-only test. Non-branching manifold has no Gast/pSom hub bifurcation, so the mutual-exclusivity correlation is not defined here.

STEP FIGURES

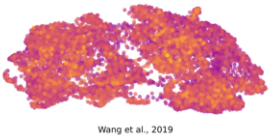
01 · Sample integration (Gate 1)



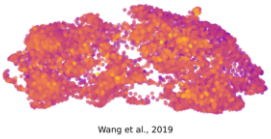
02 · Pseudotime trajectory



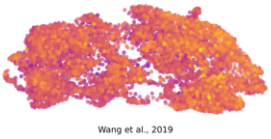
03 · Gastrulation



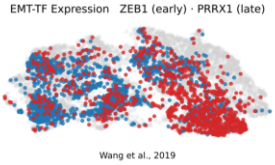
03 · Somitogenesis



03 · Neural Crest (terminal)



04 · EMT-TFs ZEB1→PRRX1



Dataset. Glioblastoma (Wang et al., 2019), 10x. 10 patients; 9,916 malignant cells, 9,634 after thresholding thin samples and removing detached islands.

Verdict. ✓ Pattern reproduces (non-branching). A single embryonic axis runs from a gastrulation/somitogenesis-high origin to a neural-crest-high terminus; pseudotime is developmental, not proliferative (cycle-score rho +0.02). EMT-TF axis ZEB1 (root) → PRRX1 (terminus).

Parameter choices

Min cells / sample	≥237 (largest multiplicative gap; keeps 7/10)
PC count	30 (chord-elbow 8 over-cuts — Chen2020 lesson)
Integration	Harmony on sample, θ=10 (mixing plateau)
Island removal	UMAP radius-graph, −16 cells / 1 island
Trajectory	scFates curve Nodes=20 λ=0.3 μ=100 (fold-free, ratio 2.87)
Root	curve tip 2 = pSom/Gast-high, Crest-low
Subtypes / annotation	3CA meta-programs (Neftel: Astro / MES-Hypoxia / OPC) + Wang 2019
Geometry	Linear-pSom/Crest (single axis; pSom strongest globally)